

Analysis of the Obstacle Potential Function

1 Potential Function

$$U_{obst} = \frac{\lambda(\Delta v / \Delta v_{max})}{k(\Delta x, \Delta y, \Delta v_x, \Delta v_y)} \quad (1)$$

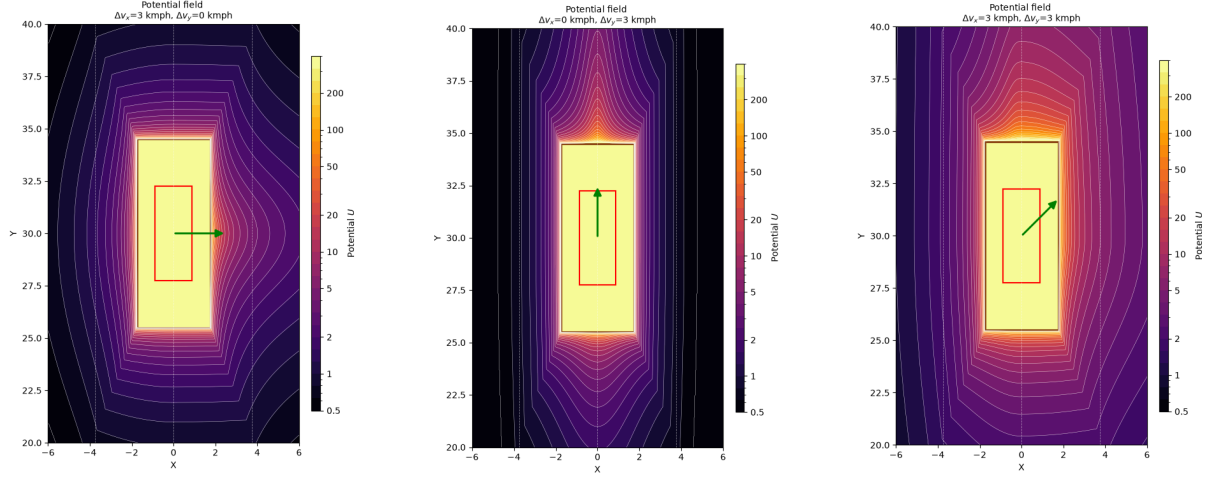
$$k(\Delta x, \Delta y, \Delta v_x, \Delta v_y) = \begin{cases} 0, & \text{if } |\Delta y| < l \text{ and } |\Delta x| < w \\ \left(1 - \frac{w}{|\Delta x|}\right) \sqrt{\left(\frac{\Delta x}{\tau_x}\right)^2 + \left(\frac{\Delta y}{\tau_y}\right)^2}, & \text{if } |\Delta y| \leq \frac{l|\Delta x|}{w} \\ \left(1 - \frac{l}{|\Delta y|}\right) \sqrt{\left(\frac{\Delta x}{\tau_x}\right)^2 + \left(\frac{\Delta y}{\tau_y}\right)^2}, & \text{otherwise} \end{cases} \quad (2)$$

$$\tau(\Delta x, \Delta v_x) = \frac{(1 + \alpha \Delta v_x) + (\alpha \Delta v_x - 1) \tanh(-\Delta x \Delta v_x)}{2} \tau_x \quad (3)$$

$$\tau(\Delta y, \Delta v_y) = \frac{(1 + \alpha \Delta v_y) + (\alpha \Delta v_y - 1) \tanh(-\Delta y \Delta v_y)}{2} \tau_y \quad (4)$$

Here $\Delta x, \Delta y$ are the relative position wrt obstacle vehicle, $\Delta v_x, \Delta v_y$ the relative velocities wrt obstacle vehicle.

1.1 Potential Field Plots



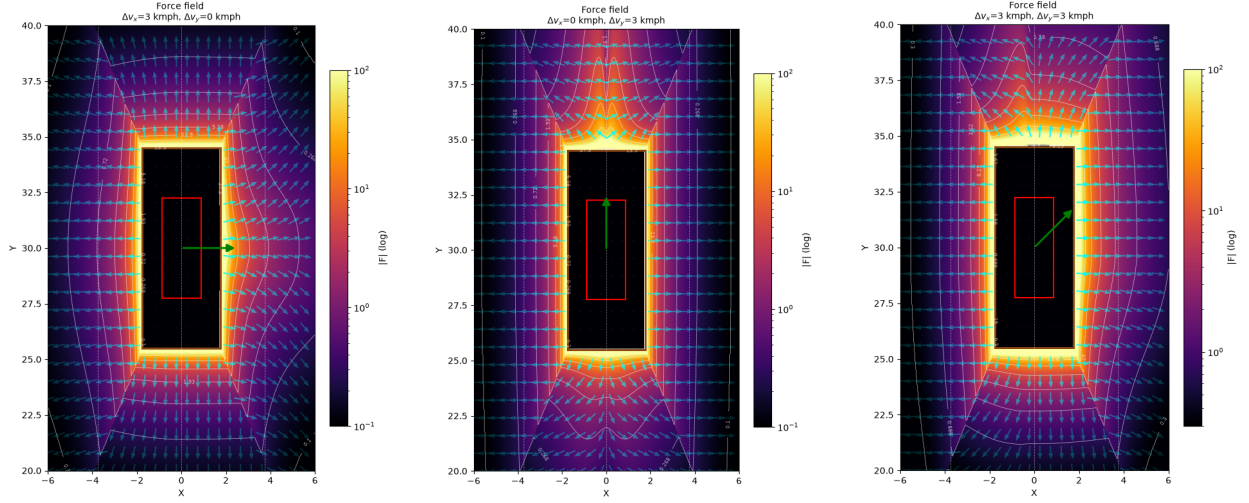
(a) Varying Δv_x

(b) Varying Δv_y

(c) Varying $\Delta v_x, \Delta v_y$

Figure 1: Obstacle potential field U_{obst} .

1.2 Force Field



(a) Varying Δv_x

(b) Varying Δv_y

(c) Varying $\Delta v_x, \Delta v_y$

Figure 2: Force field derived from U_{obst} .

1.3 Sensitivity to τ_x, τ_y

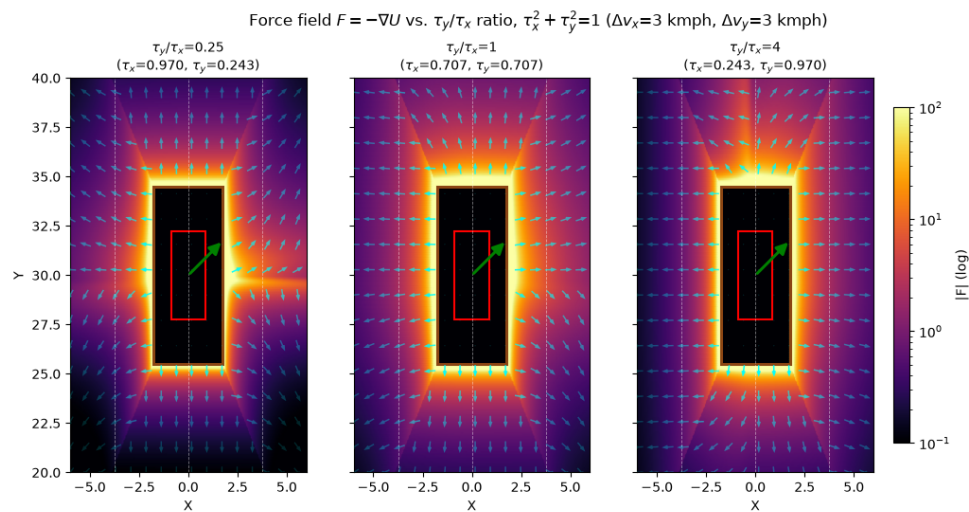


Figure 3: Sensitivity of the field to τ_x, τ_y .

1.4 Velocity at a Fixed Point

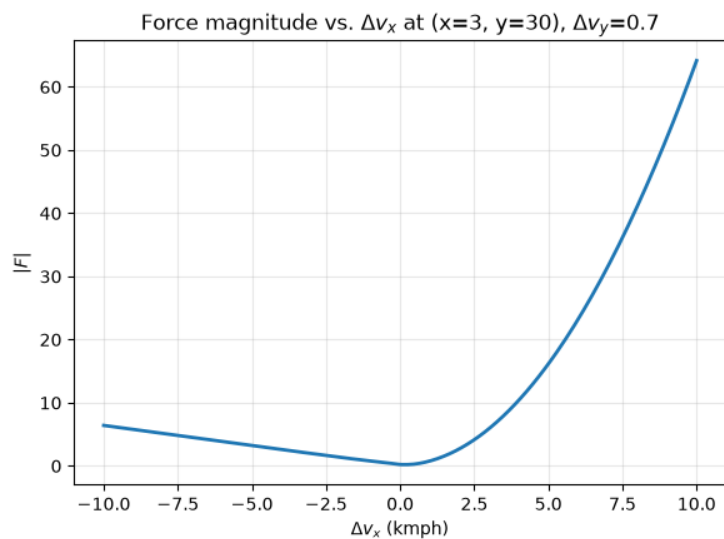


Figure 4: Velocity at a fixed point.